

Response to Reviewers

1. General Response

We thank the Associate Editor and all five reviewers for their careful and constructive evaluation. The reviews are detailed and fair, and they have substantially improved the manuscript. We are encouraged that the reviewers found the problem **important** (R1, R2, R4) and the functional-decoupling perspective **theoretically well-motivated** (R5), including gains over a strong CLIP ViT-L/14 baseline and larger improvements on hard intent categories (R1, R5).

We have organized the revision around four major concerns:

- **Fair comparison and calibration (R1).** In the Experimental Setup and Comparative Experiments, we add a matched-feature protocol in which all methods use frozen CLIP ViT-L/14 features, the same split, and validation-only threshold selection. We expand the main-results table with the controlled HLEG, LabCR, and IntCLIP results and keep their original-pipeline results explicitly separated as a robustness check. In Calibration and Threshold-free Evaluation, we add the four-cell method-versus-thresholding comparison, three-seed mean $\pm$ std, ROC/PR curves, mAP/ROC-AUC/PR-AUC, and paired bootstrap tests.
- **Novelty and theoretical depth (R2, R4).** We add the subsection “Functionally Decoupled Ambiguity Modeling” to the Method, where we give empirical support and Eq. (15) separates label-side conditional entropy from feature-side inter-class aliasing. We then add a unified-versus-decoupled table and text-teacher negative controls in the Ablation Studies to test the principle under identical features and selection rules. The Discussion now limits the claim to the measured aggregate advantage and presents the EMOTIC study as a cross-task evidence.
- **Evaluation and interpretation (R3, R4).** We expand the main-results table with zero-shot Qwen3-VL, InternVL3, and the published GPT-4o/IntentMLM result; add Fig. 4 and Fig. 3; add the per-class attribution and efficiency tables; and insert a dedicated explanation of the Easy-subset near-tie. The Discussion now records the observed class-level failure case and the scope and limitations of the second-dataset study.
- **Clarity and reproducibility (R1, R4, R5, EiC).** We rewrite the final Introduction paragraphs to state the problem gap, purpose, and contributions directly; insert Table 1 at the start of the Method; add a standalone Discussion; and expand Implementation Details with the validation-only selection boundary. We also replace “available on request” with a concrete artifact list, update the recent-work discussion and bibliography, remove the obsolete “FDIR” instances with the former appendix tables, and replace the inconsistent “Top/Bottom” wording in the Figure 2 caption with a panel-independent description of SLR-C and UTD.

Below, we provide point-by-point responses to each reviewer. Newly added or revised content is referenced by section, table, or figure. In the revised manuscript, all changes are highlighted in **red**.

2. Response to Reviewer 1

We thank the reviewer for a very constructive review and for recognizing the importance of the problem and the empirical interest of FDIL.

Comment 1

The main concerns are the fairness and interpretability of the comparisons. Please provide controlled baselines under the same backbone and thresholding protocol.

Response: This is an important fairness concern. We change both the protocol description and the result presentation. In Experimental Setup, we now define the controlled setting explicitly: FDIL, the CLIP baseline, HLEG, LabCR, and IntCLIP use the same frozen CLIP ViT-L/14 features, data split, and validation-only threshold search. In Table 2, we add lines with “(CLIP ViT-L/14)” from the methods’ original-pipeline results, which are retained only as a complementary robustness check. Under the matched protocol, FDIL leads the three representative external methods by +3.4 to +4.7 **Avg F1**. Finally, we add an Interpretability Analysis subsection and a per-class attribution table: the text identifies the largest gains in abstract, disagreement-prone intents and explicitly reports the only two regressions among the 28 classes.

Where and what we changed: Experimental Setup and Comparison with state-of-the-art methods in Sec. 4; the matched-feature rows and original-pipeline rows in Table 2; Interpretability Analysis and Table R1; the claim boundary in Sec. 5 (Limitations and Future Directions).

Table R1: Per-class F1 attribution of FDIL versus the CLIP baseline under the matched class-wise protocol. Results use a single representative seed. The table reports the five largest improvements and the only two regressions among all 28 classes.

Intent class	Subset	Baseline	FDIL	Δ
WorkILike	Hard	14.6	43.0	+28.5
PassionAboutSomething	Hard	11.3	28.6	+17.3
SocialLifeFriendship	Hard	33.9	51.2	+17.2
ManageableMakePlan	Medium	17.5	32.0	+14.5
BeatCompete	Medium	54.6	68.1	+13.5
FineDesignLearnArt-Arch	Easy	84.0	83.1	−1.0
CreativeUnique	Medium	41.9	36.6	−5.3

Table 2. Main comparison on Intentionomy test split. Avg. is the mean of Macro/Micro/Samples F1. FDIL rows report mean $\pm$ std over three seeds. [†]Quoted from IntentMLM [10].

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
Random	-	6.94	7.18	7.10	7.07
Visual [21]	CVPR 2016	22.77	30.23	28.45	27.15
Multi-label classification methods					
HiMulConE [22]	CVPR 2022	28.43	41.50	42.15	37.36
MultiGuard [23]	NeurIPS 2022	26.87	38.67	40.06	35.20
SST [24]	AAAI 2022	27.33	42.21	42.62	37.39
Query2Label [25]	-	32.12	44.64	45.15	40.64
DualCoOp [26]	NeurIPS 2022	31.06	44.42	40.40	38.63
Label noise methods					
Cleannet [27]	CVPR 2018	25.89	42.15	40.97	36.34
LS [28]	ICML 2020	24.32	42.77	43.27	36.79
Dividemix [29]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
Jia [1]	CVPR 2021	23.98	31.28	31.39	28.88
CPAD [6]	TIP 2023	27.39	40.98	41.12	36.50
HLEG [7]	TAC 2023	32.77	44.69	45.54	41.00
HLEG (CLIP ViT-L/14) [7]	TAC 2023	48.17	57.29	56.81	54.09
PIP-Net [5]	TMM 2023	32.57	45.94	47.05	41.85
LabCR [8]	TIP 2024	34.63	48.51	48.05	43.73
LabCR (CLIP ViT-L/14) [8]	TIP 2024	48.04	57.13	56.65	53.94
IntCLIP [9]	ECCV 2024	40.15	53.54	51.01	48.23
IntCLIP (CLIP ViT-L/14) [9]	ECCV 2024	47.28	56.18	54.86	52.77
IntentMLM [10]	PR 2026	43.05	54.77	56.75	51.52
Zero-shot VLM					
Qwen3-VL-8B-Instruct [30]	-	38.51	40.75	39.63	39.63
InternVL3-8B [31]	-	37.54	41.85	40.50	39.97
GPT-4o [32] [†]	-	39.48	40.57	41.45	40.50
CLIP ViT-L/14 [33]	ICML 2021	47.40	56.42	56.41	53.41
FDIL (UTD only)	Ours	50.20 $\pm$ 0.44	58.03 $\pm$ 0.11	58.79 $\pm$ 0.78	55.68 $\pm$ 0.44
FDIL	Ours	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49

Comment 2

The strong effect of class-wise thresholding.

Response: To separate calibration gain from method gain, we add a dedicated “Calibration and Threshold-free Evaluation” subsection. Its first table is a four-cell, three-seed comparison crossing method (CLIP baseline versus FDIL) with threshold type (global versus class-wise). This shows that FDIL improves Avg F1 under both class-wise (+4.08) and global (+2.31) thresholding, while class-wise calibration benefits both methods. We then add a second table with four threshold-free metrics and a 5,000-resample paired bootstrap test, plus ROC and precision–recall curves computed from the same scores in Fig. 4. FDIL improves

over the CLIP baseline on every threshold-free metric; its mean mAP gain is +4.43, and the representative-seed difference is +4.80 with 95% CI [3.58, 6.10] and $p < 0.001$.

Where and what we changed: Calibration and Threshold-free Evaluation in Sec. 4; the method $\times$ threshold cells in Table 9; the mAP, ROC-AUC, PR-AUC, confidence-interval, and significance columns in Table 10; ROC/PR curves in Fig. 4.

Table 9. Calibration decomposition under matched global/class-wise thresholding. FDIL is mean $\pm$ std over three seeds; CLIP uses a single seed.

Method	Threshold	Macro	Micro	Samples	Avg.	mAP	Hard
CLIP ViT-L/14 [33]	Global	46.22	55.05	54.51	51.92	50.67	26.84
CLIP ViT-L/14 [33]	Class-wise	47.40	56.42	56.41	53.41	50.67	28.45
FDIL	Global	47.70 $\pm$ 1.55	57.56 $\pm$ 0.41	57.43 $\pm$ 0.69	54.23 $\pm$ 0.72	55.10$\pm$0.26	30.00 $\pm$ 3.46
FDIL	Class-wise	52.13$\pm$0.81	60.06$\pm$0.24	60.28$\pm$0.42	57.49$\pm$0.49	55.10$\pm$0.26	35.02$\pm$1.77

Table 10. Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds; all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
CLIP ViT-L/14 [33]	50.67	87.04	84.90	54.89
UTD only	53.07	90.09	88.55	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	55.10	89.36	88.31	60.64

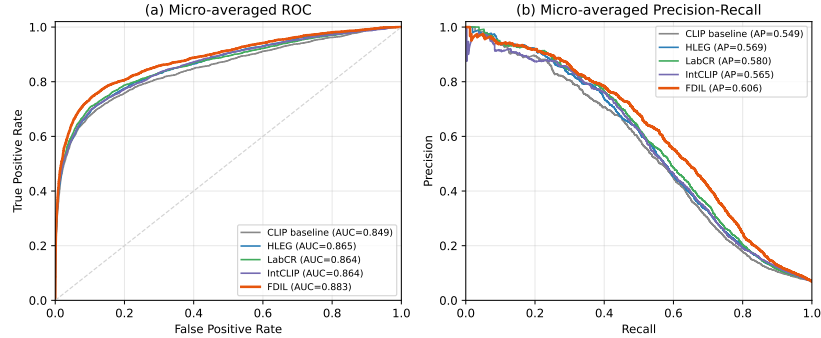


Figure 4. Micro-averaged ROC (a) and precision-recall (b) curves, comparing FDIL against the CLIP baseline and three representative methods (HLEG, LabCR, IntCLIP) reproduced under the matched CLIP ViT-L/14 features, split, and protocol.

Comment 3

Report variance over runs.

Response: Following this suggestion, we rerun the UTD-only and full FDIL settings with three seeds. We replace the former single-run entries in the main comparison with mean $\pm$ std and use the same three runs for every cell of the thresholding decomposition, so the reported variance reflects training randomness rather than different evaluation protocols.

Where and what we changed: Experimental Setup in Sec. 4 (seed protocol); the UTD and FDIL rows in Table 2; all four cells in Table 9.

Table 2. Main comparison on Intentionomy test split. Avg. is the mean of Macro/Micro/Samples F1. FDIL rows report mean $\pm$ std over three seeds. [†]Quoted from IntentMLM [10].

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
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Table 9. Calibration decomposition under matched global/class-wise thresholding. FDIL is mean $\pm$ std over three seeds; CLIP uses a single seed.

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FDIL	Class-wise	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49	55.10 $\pm$ 0.26	35.02 $\pm$ 1.77

Comment 4

Clarify validation-only model selection.

Response: We make the selection boundary explicit and audit the implementation. In the Implementation Details paragraph, we now state the sequence of operations: early stopping and hyperparameter selection use validation metrics only; global or class-wise thresholds are fitted on the validation split; those thresholds are then frozen; and the test split is evaluated once for final reporting. We also add an explicit sentence that test outputs never enter a model- or threshold-selection branch.

Where and what we changed: Implementation Details in Sec. 4, in the paragraph describing early stopping, threshold fitting, and final test evaluation.

Comment 5

The dependence on offline VLM-generated rationales and semantic priors, and incomplete reproducibility details. Please release or document generated artifacts.

Response: We now state this dependence and its cost explicitly at three locations. First, the SLR-C and UTD subsections identify which inputs are generated offline and clarify that the UTD text teacher is discarded after training, and the full FDIL model retains the lightweight semantic-prior reranking stage. Second, Implementation Details now records the reproducible pipeline order, which is feature extraction, prior construction, teacher encoding, optimization, validation-only threshold search, and agreement gating. Third, we replace “available on request” in the availability statement with an itemized release commitment covering generated priors and rationales, cached features/logits, exact prompts, fitted thresholds, seeds, checksums, and training/evaluation scripts upon acceptance.

Where and what we changed: Semantic Prior Construction and Text-derived Teacher in Sec. 3; the pipeline paragraph in Implementation Details of Sec. 4; the final Data and artifact availability statement.

Comment 6

Tone down claims about functional decoupling and SOTA performance unless the ablations isolate these factors more cleanly.

Response: We agree that these claims require evidence that separates architectural organization from backbone and calibration effects. We therefore revise the manuscript at the following specific locations.

First, in the Abstract, we replace the broad performance statement in the final sentence with protocol-qualified effect sizes: FDIL improves over representative recent methods by at least +3.4 Avg F1, +5.7 Hard F1, and +3.4 threshold-free mAP under matched frozen CLIP ViT-L/14 features and validation-only thresholding.

Second, in “Comparison with state-of-the-art methods” of Sec. 4, we add a fairness caveat immediately after Table 2. The revised paragraph explicitly states that results quoted from methods using their original, weaker backbones are not evidence of FDIL’s algorithmic superiority. We therefore add separate HLEG, LabCR, and IntCLIP rows marked “CLIP ViT-L/14” to Table 2; these rows use the same frozen features, split, and validation-only class-wise thresholds as FDIL. Under this controlled comparison, FDIL retains a +3.4 to +4.7 **Avg F1** advantage.

Third, in the Ablation Studies, we add the new subsection “Unified versus Functionally Decoupled Ambiguity Handling” and Table 6. The most direct unified control averages the semantic-prior and rationale-teacher probabilities into one distillation target and trains a single student under the same features and selection rules. It reaches 55.66 **Avg F1**, compared with 57.49 for decoupled FDIL. We also add “Negative Controls on the Text Teacher” and Table 13, where shuffled and generic rationales test whether the gain comes merely from adding external text, and gate-removal/flattening variants test the role of agreement-aware supervision.

Fourth, in “Calibration and Threshold-free Evaluation”, we add Table 9 to cross the CLIP baseline and FDIL with global versus class-wise thresholds over three seeds. We added Table 10 and Fig. 4 to report mAP, ROC-AUC, PR-AUC, micro-averaged ROC/PR curves, and a 5,000-resample paired bootstrap test. These additions show that the FDIL gain remains when class-wise thresholding is removed and when evaluation is made fully threshold-free.

Finally, in the Discussion, we add a bounded interpretation of the evidence: functional decoupling provides a measurable aggregate benefit and keeps semantic reranking and supervision stabilization separately interpretable and deployable, but the unified controls remain competitive on some metrics. Accordingly, we no longer attribute improvements observed across incompatible original pipelines to the proposed organization itself.

Where and what we changed: the final sentence of the Abstract; Comparison with state-of-the-art methods and the matched-feature rows in Table 2; Unified versus Functionally Decoupled Ambiguity Handling and Table 6; Nega-

tive Controls on the Text Teacher and Table 13; Calibration and Threshold-free Evaluation, Tables 9 and 10, and Fig. 4; the interpretation in Sec. 5.

Table 2. Main comparison on Intentionomy test split. Avg. is the mean of Macro/Micro/Samples F1. FDIL rows report mean $\pm$ std over three seeds. [†]Quoted from IntentMLM [10].

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
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Zero-shot VLM					
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GPT-4o [32] [†]	-	39.48	40.57	41.45	40.50
CLIP ViT-L/14 [33]	ICML 2021	47.40	56.42	56.41	53.41
FDIL (UTD only)	Ours	50.20 $\pm$ 0.44	58.03 $\pm$ 0.11	58.79 $\pm$ 0.78	55.68 $\pm$ 0.44
FDIL	Ours	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49

Table 6. Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
Unified SLR-C only	51.29	60.10	59.22	56.87	34.01
Unified joint-target KD	50.69	58.35	57.96	55.66	32.41
FDIL (decoupled)	52.13	60.06	60.28	57.49	35.02

Table 13. Negative controls on the UTD text teacher (Δ in points; negative denotes degradation). Full FDIL is the mean over three seeds; control deltas use a single seed.

Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD (supervised)	-4.03	-5.59	-5.17	-4.94	-2.18	-6.02
Shuffled rationale	-7.75	-12.39	-11.01	-10.40	-9.04	-7.58
Ungated teacher	-0.62	-0.41	+0.07	-0.33	+0.03	-0.69
Uniform agreement gate	-1.16	-0.93	-1.21	-1.12	-1.33	+1.13

Table 9. Calibration decomposition under matched global/class-wise thresholding. FDIL is mean $\pm$ std over three seeds; CLIP uses a single seed.

Method	Threshold	Macro	Micro	Samples	Avg.	mAP	Hard
CLIP ViT-L/14 [33]	Global	46.22	55.05	54.51	51.92	50.67	26.84
CLIP ViT-L/14 [33]	Class-wise	47.40	56.42	56.41	53.41	50.67	28.45
FDIL	Global	47.70 $\pm$ 1.55	57.56 $\pm$ 0.41	57.43 $\pm$ 0.69	54.23 $\pm$ 0.72	55.10$\pm$0.26	30.00 $\pm$ 3.46
FDIL	Class-wise	52.13$\pm$0.81	60.06$\pm$0.24	60.28$\pm$0.42	57.49$\pm$0.49	55.10$\pm$0.26	35.02$\pm$1.77

Table 10. Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds; all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
CLIP ViT-L/14 [33]	50.67	87.04	84.90	54.89
UTD only	53.07	90.09	88.55	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	55.10	89.36	88.31	60.64

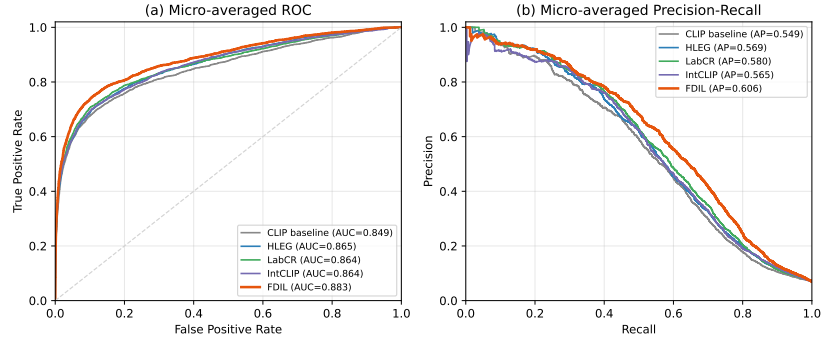


Figure 4. Micro-averaged ROC (a) and precision-recall (b) curves, comparing FDIL against the CLIP baseline and three representative methods (HLEG, LabCR, IntCLIP) reproduced under the matched CLIP ViT-L/14 features, split, and protocol.

3. Response to Reviewer 2

We thank the reviewer for acknowledging that the manuscript is well-written, technically sound, and empirically validated, and for raising the central novelty

concern.

Comment 1

The major drawback is related to insufficient novelty of the technical contribution. Most of the contributions proposed in the paper relate to previously discussed topics within vision-language modeling, semantic alignment, and uncertainty-aware learning. Although the idea of distinguishing between semantic uncertainty and supervision uncertainty is quite appealing, it does not constitute a new theoretical principle or algorithmic idea. It appears that the results are based mainly on the correct combination of existing methods rather than the development of a novel approach itself.

Response: We acknowledge that the original presentation did not distinguish the methodological contribution clearly enough from a combination of existing components. We therefore change the framing, formulation, and empirical validation together. In the Introduction, we rewrite the gap and contribution paragraphs so that the contribution is the functional separation of two measurable failure modes, rather than the mere use of a semantic prior and distillation. In the Method, we add a new subsection, “Functionally Decoupled Ambiguity Modeling.” Its diagnostic paragraph reports that the baseline Top-10 set already covers 83.9% of positive labels and retains 67.8% of its false negatives as locally recoverable candidates, indicating that much of the remaining semantic error is local ranking and calibration rather than complete retrieval failure. We then add Eq. (15), which separates supervisory ambiguity as a label-side conditional-entropy term from semantic ambiguity as a feature-side inter-class aliasing term. The text immediately following the equation maps these two terms to UTD’s supervision stabilization and SLR-C’s local ranking/calibration, making the decoupling principle explicit and testable rather than post hoc.

We test this principle in two newly added ablation blocks. “Unified versus Functionally Decoupled Ambiguity Handling” preserves the component families but changes their organization: under identical features and selection rules, merging the semantic prior and rationale teacher into one target is weaker than decoupled FDIL (55.66 **vs.** 57.49 **Avg F1**), although the strongest single-signal control remains close. “Negative Controls on the Text Teacher” replaces matched rationales with shuffled or generic text and modifies the agreement gate; shuffling rationales reduces Macro-F1 by 7.8 and Hard-F1 by 7.6, while removing or flattening the gate has smaller effects. Finally, we revise the Discussion to state the bounded conclusion: decoupling gives a modest aggregate benefit with separately measurable and deployable functions, but it is not the only competitive organization.

Where and what we changed: the rewritten gap and contribution paragraphs in Sec. 1; the new Functionally Decoupled Ambiguity Modeling subsection, diagnostic paragraph, and Eq. (15) in Sec. 3.5; the new Unified versus Functionally Decoupled Ambiguity Handling and Negative Controls subsections

with Tables 6 and 13; the bounded interpretation in Sec. 5.

Table 6. Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
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Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD (supervised)	-4.03	-5.59	-5.17	-4.94	-2.18	-6.02
Shuffled rationale	-7.75	-12.39	-11.01	-10.40	-9.04	-7.58
Ungated teacher	-0.62	-0.41	+0.07	-0.33	+0.03	-0.69
Uniform agreement gate	-1.16	-0.93	-1.21	-1.12	-1.33	+1.13

4. Response to Reviewer 3 (Associate Editor)

We thank the Associate Editor for the focused, actionable points.

Comment 1

The authors are encouraged to clearly state, in the Abstract, the quantitative improvements achieved by the proposed FDIL method compared to existing approaches.

Response: Following this recommendation, we replace the former qualitative performance statement in the final sentence of the Abstract with three controlled effect sizes: at least +3.4 **Avg F1**, +5.7 **Hard F1**, and +3.4 **threshold-free mAP** over representative recent methods. The same sentence now states the comparison boundary, matched frozen CLIP ViT-L/14 features, and validation-only thresholding, so that the gains cannot be read as comparisons across incompatible backbones or selection procedures.

Where and what we changed: the final sentence of the Abstract; the matched-feature comparison block in Table 2; threshold-free values in Table 10.

Table 2. Main comparison on Intentionomy test split. Avg. is the mean of Macro/Micro/Samples F1. FDIL rows report mean $\pm$ std over three seeds. [†]Quoted from IntentMLM [10].

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
Random	-	6.94	7.18	7.10	7.07
Visual [21]	CVPR 2016	22.77	30.23	28.45	27.15
Multi-label classification methods					
HiMulConE [22]	CVPR 2022	28.43	41.50	42.15	37.36
MultiGuard [23]	NeurIPS 2022	26.87	38.67	40.06	35.20
SST [24]	AAAI 2022	27.33	42.21	42.62	37.39
Query2Label [25]	-	32.12	44.64	45.15	40.64
DualCoOp [26]	NeurIPS 2022	31.06	44.42	40.40	38.63
Label noise methods					
Cleannet [27]	CVPR 2018	25.89	42.15	40.97	36.34
LS [28]	ICML 2020	24.32	42.77	43.27	36.79
Dividemix [29]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
Jia [1]	CVPR 2021	23.98	31.28	31.39	28.88
CPAD [6]	TIP 2023	27.39	40.98	41.12	36.50
HLEG [7]	TAC 2023	32.77	44.69	45.54	41.00
HLEG (CLIP ViT-L/14) [7]	TAC 2023	48.17	57.29	56.81	54.09
PIP-Net [5]	TMM 2023	32.57	45.94	47.05	41.85
LabCR [8]	TIP 2024	34.63	48.51	48.05	43.73
LabCR (CLIP ViT-L/14) [8]	TIP 2024	48.04	57.13	56.65	53.94
IntCLIP [9]	ECCV 2024	40.15	53.54	51.01	48.23
IntCLIP (CLIP ViT-L/14) [9]	ECCV 2024	47.28	56.18	54.86	52.77
IntentMLM [10]	PR 2026	43.05	54.77	56.75	51.52
Zero-shot VLM					
Qwen3-VL-8B-Instruct [30]	-	38.51	40.75	39.63	39.63
InternVL3-8B [31]	-	37.54	41.85	40.50	39.97
GPT-4o [32] [†]	-	39.48	40.57	41.45	40.50
CLIP ViT-L/14 [33]	ICML 2021	47.40	56.42	56.41	53.41
FDIL (UTD only)	Ours	50.20 $\pm$ 0.44	58.03 $\pm$ 0.11	58.79 $\pm$ 0.78	55.68 $\pm$ 0.44
FDIL	Ours	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49

Table 10. Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds; all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
CLIP ViT-L/14 [33]	50.67	87.04	84.90	54.89
UTD only	53.07	90.09	88.55	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	55.10	89.36	88.31	60.64

Comment 2

In Section 3.3.1, Semantic Prior Construction, the authors should provide a more detailed justification for the selected scenarios, including their representativeness and the expected outcomes in conditions not covered by the presented scenarios.

Response: We appreciate the request for an explicit selection criterion and uncovered-case behavior. We expand “Semantic Prior Construction” in three places. First, the generation paragraph now states that scenarios are not hand-picked: one fixed prompt is applied to every intent class and enumerates visually distinct realizations along five axes (subjects/objects, actions, settings, atmosphere, and photographic style), with at least three settings per intent. Second, the aggregation paragraph now explains that lexical, canonical, and scenario sources are normalized independently and averaged before they are allowed to rerank only the base model’s Top- K candidates. Third, we add an uncovered-case paragraph: if an image falls outside the generated scenarios, only the scenario match weakens; lexical and canonical priors, unchanged base logits, and the image-conditioned residual student remain active. We also complete the prior-source ablation by adding the two previously missing pairwise combinations, allowing the manuscript to show that the three sources are complementary rather than relying on a verbal assertion.

Where and what we changed: the generation-protocol, source-aggregation, and uncovered-case paragraphs in Semantic Prior Construction of Sec. 3; the added Lexical+Scenario and Canonical+Scenario rows and the three-source row in Table 7.

Table 7. Ablation of textual priors in SLR-C (reranking-only, $K=10$). Results use a single seed.

Lexical	Canonical	Scenario	Macro	Micro	Samples	Avg.	Hard
✓			49.76	59.12	58.60	55.83	31.22
	✓		49.80	59.50	58.05	55.78	29.88
		✓	49.35	58.88	57.87	55.37	32.48
✓	✓		51.18	59.39	58.11	56.23	32.92
✓		✓	50.12	59.30	58.36	55.93	32.75
	✓	✓	50.93	59.51	58.18	56.21	34.81
✓	✓	✓	50.90	59.81	58.47	56.40	32.80

Comment 3

Table 2 shows that FDIL achieves lower performance than the conventional HLEG method on the Easy subset, yet this observation is not discussed. The authors should explain the possible reasons for this result and clarify whether certain characteristics of the Easy subset contribute to the similar performance between the two methods. Further discussion of the key observations in the table is recommended.

Response: This Easy-subset exception is now stated explicitly rather than absorbed into the aggregate result. Immediately after Table 3, we add a paragraph reporting that FDIL scores 81.19, 0.30 below HLEG and slightly above the CLIP baseline (80.99). The paragraph explains that Easy classes are largely

visually unambiguous, so a strong CLIP representation already separates them well and leaves little headroom for ambiguity-targeted modules. It contrasts this near-tie with the Medium and Hard results, where FDIL’s gains over the CLIP baseline are larger. We also add a per-class attribution table and interpretation: the largest gains occur in abstract Hard classes such as *WorkILike*, *PassionAboutSomething*, and *SocialLifeFriendship*, while only one Easy class shows a small regression (-0.96 F1). The Discussion now uses this pattern to delimit the scope of the method instead of claiming uniform dominance.

Where and what we changed: the new interpretation paragraph immediately after Table 3 in Performance on Difficulty- and Content-Dependent Subsets; the new per-class analysis and Table R1; the corresponding scope statement in Sec. 5.

Table 3. Difficulty-based intent subsets (Easy/Medium/Hard); Avg. is the subset mean. FDIL is the mean over three seeds.

Method	Easy	Medium	Hard	Avg.
Random	19.86	7.11	2.81	9.93
Visual [21]	54.64	24.92	10.71	30.09
Jia (w/o hashtag) [1]	57.10	25.68	12.72	31.83
Jia (hashtag) [1]	58.86	26.30	13.11	32.76
Query2Label [25]	81.59	38.06	14.55	44.73
CPAD [6]	69.47	31.50	8.54	36.50
HLEG [7]	81.49	38.99	20.22	46.90
PIP-Net [5]	70.73	33.63	14.33	39.56
IntentMLM [10]	76.26	46.87	27.39	50.17
CLIP ViT-L/14 [33]	80.99	53.32	28.45	54.25
FDIL	81.19	57.73	35.02	57.98

Table R1. Per-class F1 attribution of FDIL versus the CLIP baseline (matched class-wise protocol, representative seed). Top: five largest improvements; bottom: the only two regressions among all 28 classes.

Intent class	Subset	Baseline	FDIL	Δ
WorkILike	Hard	14.6	43.0	+28.5
PassionAboutSomething	Hard	11.3	28.6	+17.3
SocialLifeFriendship	Hard	33.9	51.2	+17.2
ManageableMakePlan	Medium	17.5	32.0	+14.5
BeatCompete	Medium	54.6	68.1	+13.5
FineDesignLearnArt-Arch	Easy	84.0	83.1	-1.0
CreativeUnique	Medium	41.9	36.6	-5.3

5. Response to Reviewer 4

We thank the reviewer for the thorough review, which we address point by point.

Comment 1

The manuscript must require careful proofreading, such as [7] *IEEE Trans. Affect. Comput.*, and so on.

Response: Thank you for catching this presentation issue. We perform a manuscript-wide proofreading pass covering grammar, acronym consistency, table/figure references, and bibliography formatting. In the bibliography, we standardize journal and conference names and their abbreviations, including changing the flagged venue to the consistent form *IEEE Trans. Affect. Comput.*; we also check the surrounding entries for the same formatting issue.

Where and what we changed: the corrected venue field of the flagged bibliography entry and the standardized venue names throughout the Bibliography; terminology and cross-reference corrections throughout the manuscript.

Comment 2

The paper is not very clear but only claims that “A central difficulty of this task lies in two complementary forms of ambiguity: supervisory ambiguity, caused by subjective annotation over plausible intentions, and semantic ambiguity, caused by semantically adjacent and overlapping intents.” Please give more analysis about these disadvantages in the existing visual intention understanding.

Response: We appreciate this request for a sharper diagnosis. In the Introduction, we expand the paragraph that introduces the two ambiguities with concrete consequences: semantically adjacent intents create local class confusion, while subjective multi-label annotation makes a missing label an unreliable negative. In Related Work, we reorganize the closing comparison to show that structure-aware and semantic-prior methods mainly address class-side overlap, whereas robust-loss and disagreement-aware methods mainly address supervision-side uncertainty; treating either as the sole source leaves the other failure mode unresolved. We then add empirical and theoretical support in the new Method subsection “Functionally Decoupled Ambiguity Modeling.” Its diagnostic paragraph shows that the baseline Top-10 set already covers 83.9% of positive labels and retains 67.8% of false negatives as locally recoverable candidates, locating much of the remaining semantic error in local ranking and calibration. Eq. (15) separately represents label-side conditional entropy and feature-side inter-class aliasing, and the following paragraph maps them to UTD and SLR-C. Thus, the

revision now moves from problem examples to gaps in existing method families, to a measured diagnostic, and a formal argument.

Where and what we changed: the expanded ambiguity-diagnosis paragraph in Sec. 1; the closing comparison in Sec. 2; the diagnostic paragraph, Eq. (15), and module mapping in Sec. 3.5.

Comment 3

The clear purpose of work the proposed work is lacking in the introduction section.

Response: We rewrite the final transition and contribution paragraphs of the Introduction. The transition now states the unresolved problem that existing methods do not jointly handle class-side semantic overlap and label-side supervisory uncertainty, and then states the purpose directly: FDIL converts external semantic knowledge into a structured prior and an uncertainty-gated training signal that targets the two ambiguities separately. We also revise the contribution bullets so that each names a function (local semantic reranking/calibration, uncertainty-guided distillation, and controlled validation) rather than merely listing model components.

Where and what we changed: the final problem-to-method transition and the three contribution paragraphs in Sec. 1.

Comment 4

The introduction should be re-discussed and extended. I think that more Large Language model-based visual intention understanding methods should be mentioned. In the introduction, this section is hard for readers to understand its latest development.

Response: To bring the literature discussion up to date, we add IntCLIP (ECCV 2024) and IntentMLM (Pattern Recognition 2026) to both the motivation and the Image Intent Recognition review. Rather than appending a citation list, the new Related Work text describes their distinct roles: IntCLIP adapts CLIP for intent recognition, whereas IntentMLM uses an MLLM for prompt-based intent reasoning. We then add a short comparison paragraph explaining that MLLMs can supply broad semantic knowledge, but do not by themselves provide dataset-specific calibration or agreement-aware supervision. Finally, the experiment section now contains a dedicated “Zero-shot Multimodal Large Model Comparison” and new Qwen3-VL, InternVL3, and published GPT-4o/IntentMLM rows in the main table, linking the literature update to measured evidence.

Where and what we changed: the recent-method motivation in Sec. 1; the added IntCLIP, IntentMLM, and MLLM comparison paragraphs in Image

Intent Recognition of Sec. 2; Zero-shot Multimodal Large Model Comparison and its rows in Table 2.

Table 2. Main comparison on Intentionomy test split. Avg. is the mean of Macro/Micro/Samples F1. FDIL rows report mean $\pm$ std over three seeds. [†]Quoted from IntentMLM [10].

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
Random	-	6.94	7.18	7.10	7.07
Visual [21]	CVPR 2016	22.77	30.23	28.45	27.15
Multi-label classification methods					
HiMulConE [22]	CVPR 2022	28.43	41.50	42.15	37.36
MultiGuard [23]	NeurIPS 2022	26.87	38.67	40.06	35.20
SST [24]	AAAI 2022	27.33	42.21	42.62	37.39
Query2Label [25]	-	32.12	44.64	45.15	40.64
DualCoOp [26]	NeurIPS 2022	31.06	44.42	40.40	38.63
Label noise methods					
Cleannet [27]	CVPR 2018	25.89	42.15	40.97	36.34
LS [28]	ICML 2020	24.32	42.77	43.27	36.79
Dividemix [29]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
Jia [1]	CVPR 2021	23.98	31.28	31.39	28.88
CPAD [6]	TIP 2023	27.39	40.98	41.12	36.50
HLEG [7]	TAC 2023	32.77	44.69	45.54	41.00
HLEG (CLIP ViT-L/14) [7]	TAC 2023	48.17	57.29	56.81	54.09
PIP-Net [5]	TMM 2023	32.57	45.94	47.05	41.85
LabCR [8]	TIP 2024	34.63	48.51	48.05	43.73
LabCR (CLIP ViT-L/14) [8]	TIP 2024	48.04	57.13	56.65	53.94
IntCLIP [9]	ECCV 2024	40.15	53.54	51.01	48.23
IntCLIP (CLIP ViT-L/14) [9]	ECCV 2024	47.28	56.18	54.86	52.77
IntentMLM [10]	PR 2026	43.05	54.77	56.75	51.52
Zero-shot VLM					
Qwen3-VL-8B-Instruct [30]	-	38.51	40.75	39.63	39.63
InternVL3-8B [31]	-	37.54	41.85	40.50	39.97
GPT-4o [32] [†]	-	39.48	40.57	41.45	40.50
CLIP ViT-L/14 [33]	ICML 2021	47.40	56.42	56.41	53.41
FDIL (UTD only)	Ours	50.20 $\pm$ 0.44	58.03 $\pm$ 0.11	58.79 $\pm$ 0.78	55.68 $\pm$ 0.44
FDIL	Ours	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49

Comment 5

You may consider including a table of notations.

Response: We follow this suggestion by inserting a compact notation table immediately after the Method overview, before the formal problem definition. It defines the image and label variables, base logits and candidate set, semantic priors, rationale teacher/student outputs, agreement score, gate, and loss weights used later in the equations. We also check the subsequent equations

against the table and retain local definitions only for one-off symbols. To remain within the 35-page limit, the table covers the notation needed for the model formulation, ambiguity decomposition, and training objective without duplicating every locally used index.

Where and what we changed: the new Table 1 between Overview and Problem Formulation in Sec. 3, with corresponding symbol usage checked in the subsequent SLR-C, UTD, and risk-decomposition equations.

Table 1. Key notation used in the method.

Symbol	Meaning
x	input image
$\mathcal{Y}, C$	intent label space and number of classes
$\mathbf{y}^{\text{soft}}, \mathbf{y}^{\text{hard}}$	soft annotation / binarized hard labels
$\omega \in (0, 1]$	sample-level annotator agreement score
$\phi = f_{\text{img}}(x)$	frozen visual image feature
$\hat{s}_c^{\text{prior}}$	normalized heterogeneous semantic prior
$\mathcal{C}_K(x), K$	Top- K candidate set and its size
s_c^{full}	final refined score for class c
t_c	class-wise decision threshold (validation-only)
$\mathbf{p}^{\text{tea}}, \tau$	text-teacher distribution and temperature
$g = 1 - \omega$	teacher-branch gate weight
$\mathcal{L}_{\text{total}}$	total training loss

Comment 6

There should be larger-scale datasets to better evaluate the proposed methods.

Response: We share the reviewer’s view that evidence beyond Intentionomy is useful for assessing scope. We add a new cross-dataset paragraph to “Limitations and Future Directions” and report the full auxiliary results here. Specifically, we construct a controlled 5×5 cross-validation study on the multi-annotated EMOTIC validation and test pool (10,613 persons), using out-of-fold teacher predictions so that privileged distillation is evaluated without target leakage. Across 25 seed×fold runs, both standard and agreement-gated distillation improve the baseline on all four aggregate metrics ($p < 0.01$ in the paired fold-level tests), while the gated variant is comparable to standard distillation overall.

The agreement-stratified results below provide a more informative view of the gate. On the lowest-agreement (1/3) subset, gated KD improves the baseline by +0.43 **Macro-F1**, +0.69 **Micro-F1**, and +0.69 **Samples-F1** (all $p < 0.01$), and is numerically strongest among the two distillation variants on Macro-, Micro-, and Hard-F1. At agreement 2/3, it is strongest on Macro-F1, mAP, and Hard-F1. In contrast, no reliable gated gain appears on fully agreed samples, where the gate is designed to defer to direct supervision. We interpret this result conservatively: privileged semantic supervision transfers to EMOTIC, and

agreement conditioning concentrates its useful behavior on uncertain samples, while Intentionomy remains the dataset that establishes the gate’s incremental advantage with native training-time agreement. We add this cross-dataset evidence and its limitations to the Discussion, showing that the framework has a degree of generalization beyond visual intention recognition without presenting EMOTIC as a new benchmark result.

Table R2: Results on the multi-annotated EMOTIC validation and test pool. Controlled 5×5 cross-validation, mean over 25 seed $\times$ fold runs. $*p < 0.01$, paired t -test vs. baseline.

Method	Macro	Micro	Samples	mAP
baseline	43.34	60.32	60.01	42.57
+ UTD(w/o gated)	43.68*	60.95*	60.75*	42.77*
+ UTD(with gated)	43.72*	60.97*	60.67*	42.77*

Table R3: Agreement-stratified EMOTIC results over the same 25 seed $\times$ fold runs. Values are percentages; bold marks the best image-only student in each agreement group.

Agreement	Method	Macro	Micro	Samples	mAP	Hard
1/3	Baseline	44.15	62.19	62.03	44.15	36.92
	+ UTD(w/o gated)	44.55	62.83	62.75	44.37	37.38
	+ UTD(with gated)	44.58	62.88	62.72	44.32	37.45
2/3	Baseline	40.84	54.39	54.84	44.63	34.30
	+ UTD(w/o gated)	41.04	55.00	55.63	44.85	34.44
	+ UTD(with gated)	41.20	54.92	55.41	45.18	34.69
1	Baseline	22.42	36.31	36.60	34.07	18.23
	+ UTD(w/o gated)	22.83	37.28	37.80	34.07	18.05
	+ UTD(with gated)	22.15	36.79	37.20	34.34	17.41

Where and what we changed: the new cross-dataset scope-and-limitations paragraph in Sec. 5 (Limitations and Future Directions); the aggregate 5×5 protocol and results in Table R2 of this letter; the agreement-stratified analysis in Table R3 of this letter.



Comment 7

Experimental evaluation is not sufficient. More comparisons with state-of-the-art Large Language Model methods (such as 2025 and 2026) and t-SNE are needed.

Response: In response, we created a dedicated “Zero-shot Multimodal Large Model Comparison” subsection and added zero-shot Qwen3-VL-8B-Instruct and InternVL3-8B under the same Intentionomy label set and output protocol, together with the published GPT-4o result from IntentMLM. Their Macro-, Micro-, Samples-, Avg-, and difficulty-stratified F1 values are now separate rows in Table 2. Because all three prompt-only models remain below the task-trained CLIP/FDIL models, the accompanying paragraph now interprets MLLMs as semantic sources rather than substitutes for task-specific learning. We also added an “Interpretability Analysis” subsection and Fig. 3, which compares the CLIP baseline and FDIL embeddings for selected semantically adjacent intents and explains the observed change in local separation.

Where and what we changed: the new Zero-shot Multimodal Large Model Comparison subsection and Qwen3-VL, InternVL3, and GPT-4o rows in Table 2; the new Interpretability Analysis subsection and t-SNE visualization in Fig. 3.

Comment 8

Please use some ROC-based curves for a clear evaluation of the proposed method with other state-of-the-art methods.

Response: We add the requested curves and the corresponding numerical checks in “Calibration and Threshold-free Evaluation.” Fig. 4 now plots micro-averaged ROC and precision-recall curves for FDIL, the CLIP baseline, HLEG, LabCR, and IntCLIP, all computed from the same matched CLIP ViT-L/14 features and split. Table 10 is added beside this analysis to report ROC-AUC, PR-AUC, mAP, bootstrap confidence intervals, and paired tests rather than

Table 2. Main comparison on Intentionomy test split. Avg. is the mean of Macro/Micro/Samples F1. FDIL rows report mean $\pm$ std over three seeds. [†]Quoted from IntentMLM [10].

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
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MultiGuard [23]	NeurIPS 2022	26.87	38.67	40.06	35.20
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Query2Label [25]	-	32.12	44.64	45.15	40.64
DualCoOp [26]	NeurIPS 2022	31.06	44.42	40.40	38.63
Label noise methods					
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Dividemix [29]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
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CPAD [6]	TIP 2023	27.39	40.98	41.12	36.50
HLEG [7]	TAC 2023	32.77	44.69	45.54	41.00
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LabCR [8]	TIP 2024	34.63	48.51	48.05	43.73
LabCR (CLIP ViT-L/14) [8]	TIP 2024	48.04	57.13	56.65	53.94
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IntCLIP (CLIP ViT-L/14) [9]	ECCV 2024	47.28	56.18	54.86	52.77
IntentMLM [10]	PR 2026	43.05	54.77	56.75	51.52
Zero-shot VLM					
Qwen3-VL-8B-Instruct [30]	-	38.51	40.75	39.63	39.63
InternVL3-8B [31]	-	37.54	41.85	40.50	39.97
GPT-4o [32] [†]	-	39.48	40.57	41.45	40.50
CLIP ViT-L/14 [33]	ICML 2021	47.40	56.42	56.41	53.41
FDIL (UTD only)	Ours	50.20 $\pm$ 0.44	58.03 $\pm$ 0.11	58.79 $\pm$ 0.78	55.68 $\pm$ 0.44
FDIL	Ours	52.13 $\pm$ 0.81	60.06 $\pm$ 0.24	60.28 $\pm$ 0.42	57.49 $\pm$ 0.49

asking the reader to judge curve shape alone. The paragraph following the figure explicitly states that FDIL is higher on both curves while keeping the comparison within the controlled setting.

Where and what we changed: the new Calibration and Threshold-free Evaluation subsection in Sec. 4; micro-averaged ROC/PR curves in Fig. 4; ROC-AUC, PR-AUC, mAP, confidence-interval, and paired-test columns in Table 10.

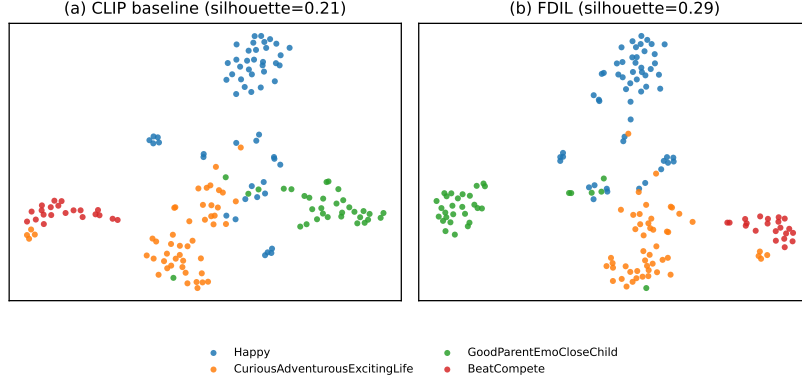


Figure 3. t-SNE of the decision-score space on four semantically adjacent intents. FDIL yields tighter, better-separated same-intent clusters than the CLIP baseline.

Table 10. Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds; all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
CLIP ViT-L/14 [33]	50.67	87.04	84.90	54.89
UTD only	53.07	90.09	88.55	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	55.10	89.36	88.31	60.64

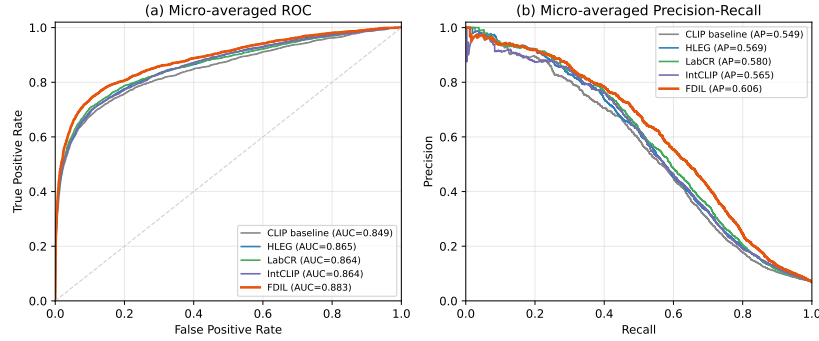


Figure 4. Micro-averaged ROC (a) and precision-recall (b) curves, comparing FDIL against the CLIP baseline and three representative methods (HLEG, LabCR, IntCLIP) reproduced under the matched CLIP ViT-L/14 features, split, and protocol.

Comment 9

A new section called “Discussion” could be added. The authors can discuss the main findings and also the important advantages and disadvantages of the proposed model.

Response: Following this suggestion, we insert a standalone Discussion between Experiments and Conclusion. Its first subsection, “A Functionally Decoupled View of Ambiguity,” summarizes the candidate-recall evidence, the role of SLR-C and UTD, and the controlled unified-versus-decoupled result. Its second subsection, “Limitations and Future Directions,” records the limited headroom on visually unambiguous Easy classes, dependence on offline semantic artifacts, and the bounded interpretation of the EMOTIC evidence. We also shorten the Conclusion so that it states the takeaway without duplicating this analysis.

Where and what we changed: the new Sec. 5, specifically A Functionally Decoupled View of Ambiguity and Limitations and Future Directions; the shortened Sec. 6.

Comment 10

It would add more value to the paper if more theoretical analysis were conducted, e.g., from information-theoretical or feature selection perspectives.

Response: We add a new Method subsection, “Functionally Decoupled Ambiguity Modeling,” rather than placing theory only in the Discussion. The subsection first reports the candidate-recall diagnostic and casts SLR-C as local candidate selection and reranking: because most positives are already retrievable within the Top-10 set, the principal residual problem is separating neighboring intents. It then introduces Eq. (15), an excess-risk decomposition that separates label-side conditional entropy from feature-side inter-class aliasing, and maps the two terms to UTD and SLR-C. To test the implications, we add a unified-versus-decoupled table, text-teacher negative controls, and threshold-free evaluation; the detailed numbers are given in our response to Reviewer 2’s novelty concern rather than duplicated here.

Where and what we changed: the diagnostic, feature-selection interpretation, Eq. (15), and module mapping in Sec. 3.5; the new controlled tests in Tables 6, 13, and 10.

Table 6. Unified versus functionally decoupled ambiguity handling. Results are means over three seeds.

Method	Macro	Micro	Samples	Avg.	Hard
Unified UTD only	50.20	58.03	58.79	55.68	30.27
Unified SLR-C only	51.29	60.10	59.22	56.87	34.01
Unified joint-target KD	50.69	58.35	57.96	55.66	32.41
FDIL (decoupled)	52.13	60.06	60.28	57.49	35.02

Table 13. Negative controls on the UTD text teacher (Δ in points; negative denotes degradation). Full FDIL is the mean over three seeds; control deltas use a single seed.

Condition	Macro	Micro	Samples	Avg.	mAP	Hard
FDIL (full)	52.13	60.06	60.28	57.49	55.10	35.02
w/o UTD (supervised)	-4.03	-5.59	-5.17	-4.94	-2.18	-6.02
Shuffled rationale	-7.75	-12.39	-11.01	-10.40	-9.04	-7.58
Ungated teacher	-0.62	-0.41	+0.07	-0.33	+0.03	-0.69
Uniform agreement gate	-1.16	-0.93	-1.21	-1.12	-1.33	+1.13

Table 10. Threshold-free, rank-based metrics. mAP is the macro average precision. FDIL/UTD mAP values are means over three seeds; all AUCs and CLIP mAP use a single representative seed.

Method	mAP	Macro ROC-AUC	Micro ROC-AUC	Micro PR-AUC
CLIP ViT-L/14 [33]	50.67	87.04	84.90	54.89
UTD only	53.07	90.09	88.55	59.59
SLR-C only	53.79	88.24	86.73	58.60
FDIL	55.10	89.36	88.31	60.64

Comment 11

A comparative analysis of model parameters, FLOPs, or inference times is necessary.

Response: We add a separate “Efficiency Comparison” subsection and a method-by-method table. For FDIL and each feature-level baseline, Table 15 now reports trainable parameters, FLOPs, and per-image latency measured in the same environment over shared frozen CLIP features. The accompanying paragraph identifies which FDIL components are used only during training and which remain at inference, and reports the resulting cost: 2.41M trainable parameters, 4.93M FLOPs, and 0.47ms per image. We list IntentMLM separately and explain why an end-to-end MLLM pipeline is not directly comparable to a feature-level head.

Where and what we changed: the new Efficiency Comparison subsection in Sec. 4; the Params, FLOPs, and Latency columns in Table 15; the paragraph distinguishing training-only and inference-time components.

6. Response to Reviewer 5

We thank the reviewer for the positive assessment of the decoupling perspective and the module design, and for the precise minor corrections.

Table 15. Feature-level inference cost on frozen CLIP ViT-L/14 features: params (inference-active heads), per-image FLOPs, and batch-1 latency on an RTX 4090.

Method	Params (M)	FLOPs (M)	Latency (ms)
Baseline	1.20	2.40	0.109
FDIL (UTD only)	1.20	2.40	0.109
FDIL	2.41	4.93	0.466
HLEG [7]	203.70	12 075.9	2.725
IntCLIP [9]	0.02	22.02	0.111
IntentMLM [10]	MLLM: $\sim 10^9$ params, 10–100 ms/image		

Comment 1

Figure 2 caption inconsistency: The caption refers to “(Top) UTD” and “(Bottom) SLR-C”, but the figure shows “A” as the top and “B” as the bottom. Please revise the caption or the figure to ensure consistent correspondence.

Response: Thank you for identifying this inconsistency. To satisfy the page limit, we shorten the Figure 2 caption and remove the former “Top/Bottom” panel description. The revised caption now summarizes the functions of SLR-C and UTD without making the inconsistent correspondence.

Where and what we changed: the rewritten panel-independent caption of Fig. 2.

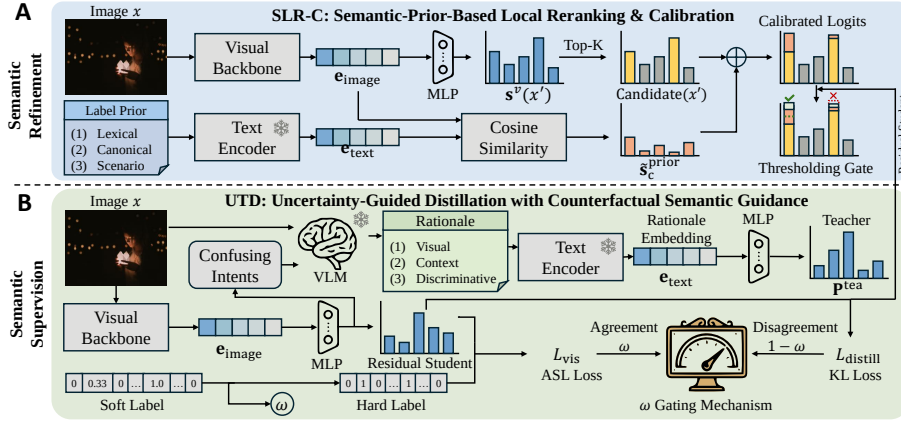


Figure 2. Overview of FDIL. SLR-C constructs a textual semantic prior and performs local reranking plus residual refinement. UTD uses a rationale-based text teacher and an agreement-aware gate to stabilize learning under ambiguous supervision.

Comment 2

Missing experimental settings in Table 5:: Two additional configurations should be included: (i) Lexical + Scenario, and (ii) Canonical + Scenario.

Response: We add the two rows to the prior-source ablation in Table 7 and rerun them under the same $K=10$ reranking-only protocol as every existing row. We also revise the paragraph below the table so that it compares all three pairwise combinations: both new pairs improve over their single-source components, Canonical+Scenario is the strongest pair on Hard-F1, and the three-source ensemble remains best on aggregate Avg-F1. Thus, the revision changes both the table coverage and the interpretation rather than simply inserting two numbers.

Where and what we changed: the added Lexical+Scenario and Canonical+Scenario rows in Table 7; the revised comparison paragraph immediately following the table in Heterogeneous Priors in SLR-C.

Table 7. Ablation of textual priors in SLR-C (reranking-only, $K=10$). Results use a single seed.

Lexical	Canonical	Scenario	Macro	Micro	Samples	Avg.	Hard
✓			49.76	59.12	58.60	55.83	31.22
	✓		49.80	59.50	58.05	55.78	29.88
		✓	49.35	58.88	57.87	55.37	32.48
✓	✓		51.18	59.39	58.11	56.23	32.92
✓		✓	50.12	59.30	58.36	55.93	32.75
	✓	✓	50.93	59.51	58.18	56.21	34.81
✓	✓	✓	50.90	59.81	58.47	56.40	32.80

Comment 3

Unexplained anomaly in Table 6: Provide a justification for why $K = 5$ $K = 28$ yield better performance than $K = 10$.

Response: We revisit this anomaly by replacing the former single-seed comparison with a three-seed sweep of $K \in \{5, 10, 28\}$ under the final protocol. Table 8 now reports mean±std for the complete metric set and identifies $K = 10$ as the best aggregate setting, so the earlier single-seed dip is no longer used in our conclusion. We rewrite the accompanying interpretation to explain both sides of the trade-off: $K = 5$ can exclude plausible positive labels, whereas $K = 28$ admits all classes and introduces distractors without improving semantic coverage; $K = 10$ best balances candidate recall and local separability. We also change the default setting stated in the Method and Implementation Details to $K = 10$ for consistency.

Where and what we changed: the replaced three-seed results and mean±std entries in Table 8; the rewritten trade-off paragraph in Sensitivity to Candidate

Size K ; the consistent $K = 10$ setting in Semantic Local Reranking and Implementation Details.

Table 8. Sensitivity to candidate size K (class-wise thresholding). Results are means over three seeds.

Top- K	Macro	Micro	Samples	Avg.	Hard
5	51.19	59.03	59.42	56.54	31.60
10	52.13	60.06	60.28	57.49	35.02
28	51.64	59.59	59.71	56.98	34.02

Comment 4

Typo in Appendix tables: In Table B.1 and Table B.2 (page 35), “FDIR” should be corrected to “FDIL”.

Response: Thank you for identifying this typo. During revision, the former Appendix Tables B.1–B.2 are removed because their validation-split results were redundant with the revised main analysis, and the manuscript had to remain within 35 pages; consequently, the two reported “FDIR” instances are no longer present. We nevertheless perform a manuscript-wide search across the text, captions, and tables, and correct or confirm every remaining acronym occurrence as “FDIL.”

Where and what we changed: removal of the former Appendix Tables B.1–B.2; acronym consistency checked throughout the main text, table entries, and captions.

7. Response to Editor-in-Chief Checklist

We confirm that the revision adheres to the editorial checklist below.

Comment 1

Take a careful look at your bibliography and they cover the state of the art. Missing references from last and current year most probably would mean you are missing the state of the art and the revision process can be delayed being asked to update it. Please do not make excessive citation to arXiv papers, but substitute them with their peer-reviewed versions, or papers from a single conference series. Do not cite large groups of papers without individually commenting on them. So we discourage “In prior work [1,2,3,4,5,6] ...”. Your bibliography in the final version after the revision still should be between 35–55 items.

Response: Following this guidance, we audit both the references and how they are discussed. We add recent peer-reviewed work, including IntCLIP (ECCV 2024) and IntentMLM (Pattern Recognition 2026), replace preprint records with published versions where available, and standardize venue metadata. In the Introduction and Related Work, we break up broad citation clusters and add method-specific sentences describing what the most relevant recent approaches contribute and how they differ from FDIL. The bibliography now contains **38 references**, within the requested range.

Where and what we changed: the recent-work motivation in Sec. 1; the method-specific IntCLIP, IntentMLM, uncertainty-learning, and MLLM discussion in Sec. 2; updated publication records and venue fields in the Bibliography.

Comment 2

Please make sure the revised version is relevant to the readership of the Pattern Recognition field. To this end, please make sure you cite RECENT work from the field of pattern recognition not only the Pattern Recognition journal.

Response: We add **four references** from the broader pattern-recognition and multimodal-recognition literature: EMOTIC for context-aware visual affect recognition, GPT-4o as a recent multimodal model, and the recent open multimodal models Qwen3-VL and InternVL3. These additions complement the visual-intent references already discussed in the manuscript, including IntCLIP (ECCV 2024) and IntentMLM (Pattern Recognition 2026), and broaden the coverage beyond papers published specifically in the *Pattern Recognition* journal. We cite and discuss these works where they directly support the cross-dataset evaluation and the comparison with recent multimodal recognition models, rather than adding them as an undifferentiated citation list.

Where and what we changed: the added EMOTIC reference in Sec. 5; the added GPT-4o, Qwen3-VL, and InternVL3 references in Zero-shot Multimodal Large Model Comparison of Sec. 4; the corresponding four new entries in the Bibliography.

Comment 3

Although the revision could lead to extending your article, it still can not exceed the page limits or violate the format, i.e. double spaced SINGLE column with a maximum of 35 pages for a regular paper and 40 pages for a review.

Response: The revised article now follows the required double-spaced, single-column format and is exactly **35 pages**. To accommodate the requested additions, we integrate the notation table directly after the Method overview, combine related interpretability results into one subsection, consolidate the UTD

ablations, and remove redundant validation-split appendix tables. Exact generation prompts and large auxiliary outputs are identified in the artifact-release statement instead of being reproduced in the manuscript. The newly requested EMOTIC detail is summarized in the Discussion and reported fully in this response letter. These changes preserve every reviewer-requested conclusion while keeping the article within the formal limit.

Where and what we changed: the full revised manuscript in double-spaced, single-column format (35 pages); Table 1 after the Method overview; consolidated Interpretability Analysis and UTD Ablations in Sec. 4; removal of former Appendix Tables B.1–B.2; the final Data and artifact availability statement.

Table 1. Key notation used in the method.

Symbol	Meaning
x	input image
$\mathcal{Y}, C$	intent label space and number of classes
$\mathbf{y}^{\text{soft}}, \mathbf{y}^{\text{hard}}$	soft annotation / binarized hard labels
$\omega \in (0, 1]$	sample-level annotator agreement score
$\phi = f_{\text{img}}(x)$	frozen visual image feature
$\tilde{s}_c^{\text{prior}}$	normalized heterogeneous semantic prior
$\mathcal{C}_K(x), K$	Top- K candidate set and its size
s_c^{full}	final refined score for class c
t_c	class-wise decision threshold (validation-only)
$\mathbf{p}^{\text{tea}}, \tau$	text-teacher distribution and temperature
$g = 1 - \omega$	teacher-branch gate weight
$\mathcal{L}_{\text{total}}$	total training loss

8. Closing Statement

We believe the revision substantially strengthens the novelty framing, the fairness and interpretability of the evaluation, and the reproducibility of the work. We thank the Associate Editor and reviewers again for their careful guidance and hope that the revised manuscript now addresses all concerns.